

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2022-2023
FULL MARKS: 75

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses. The symbols have their usual meanings.

1. The game tree for a game between Aafiya and Bashira is shown in Figure 1:

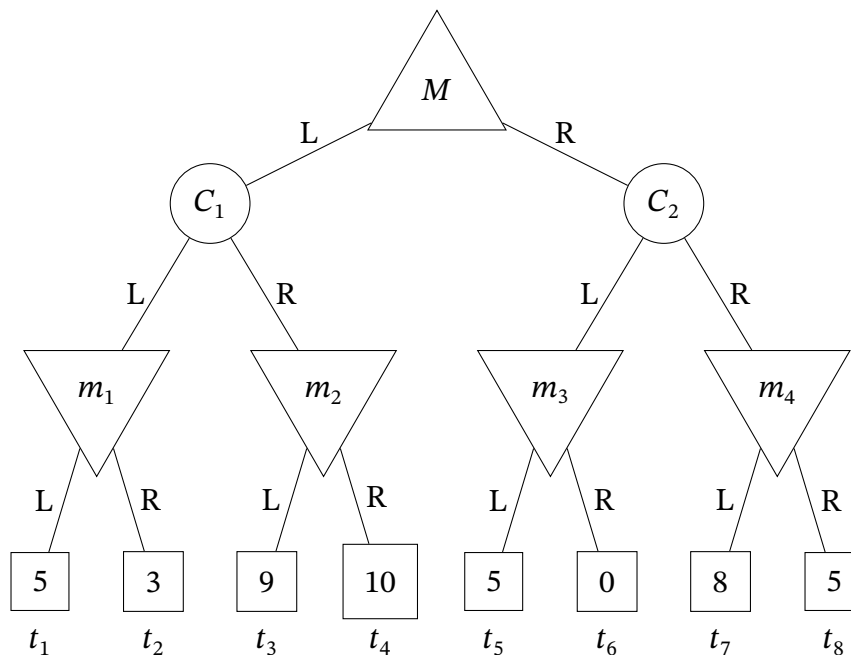


Figure 1: Game Tree for Question 1

Here, L indicates the action of going left and R indicates the action of going right. For the chance nodes, the probability of taking the action 'L' is, $P = 0.8$. The values of the terminal nodes will be between 0 and 10 (inclusive). Aafiya moves first.

- a) Recommend the optimal action for Aafiya.

8
(CO3)
(PO3)

Solution:

$m_1 = 3$
 $m_2 = 9$
 $m_3 = 0$
 $m_4 = 5$
 $C_1 = 4.2$
 $C_2 = 1$
 $M = 4.2$

Aafiya will go left.

Rubric:

- 1 points for each value
- 1 point for the decision

- b) Assume that we update the value of t_5 to 1. Determine the value for P such that Aafiya takes the same action that you recommended in Question 1.a).

5
(CO3)
(PO3)

Solution:

No change is required, since Aafiya is already taking the left action and reducing the value of t_5 only decreases that value achieved by taking the right action.

[Other valid values/range for P is also treated as correct answer] **Rubric:**

- 3 points for the decision
- 2 points for justification

- c) “Even though the given game tree corresponds to an Expectimax problem, some leaf nodes can be pruned using Alpha-Beta Pruning” - Do you agree with the statement? If you do, with a brief explanation, identify the nodes that can be pruned. If you do not, briefly explain your reasoning.

1 + 4
(CO1)
(PO1)

Solution:

Yes, I agree.

Once we have established that C_1 is equal to 4.2, and we have the values $t_5 = 5$ and $t_6 = 0$, we can deduce that the value of m_3 is 0. Consequently, regardless of the values assigned to t_7 and t_8 , the highest possible value for C_2 remains at 2, which is less than C_1 . Hence, it is possible to prune t_7 and t_8 .

Rubric:

- 1 point for correct decision
- 2 points for explanation
- 1 point for each correct node pruning

- d) Consider the general case where the terminal nodes can take on any real value between 0 and 10 (inclusive) instead of the given values. Does applying Alpha-Beta Pruning always result in the same outcome as of Question 1.c)? Justify your position with appropriate reasoning and examples.

1 + 4
(CO1)
(PO1)

Solution:

No.

Since the terminal values are no longer fixed, the nodes pruned may change. In fact, given suitable circumstance, nodes m_4 , t_6 , t_7 , and t_8 can be pruned. Since we do not prune on equality, t_1 , t_2 , t_3 , and t_4 should not be pruned in order to determine the value of C_1 . Then, if t_5 is way too small, we do not need to look at the remaining nodes. For example, $t_1 = 10$, $t_2 = 10$, $t_3 = 10$, $t_4 = 10$, and $t_5 = 0$, we know that $C_1 = 10$ and the maximum value of $C_2 = 2$.

Rubric:

- 1 point for correct decision
- 2 points for justification
- 2 points for example

2. Assume that six friends Rifat (R), Atiq (A), Farhan (F), Ishmam (I), Tabassum (T), and Sabrina (S) are standing in a queue, each occupying a unique spot among the six possible spots labeled 1 to 6. Farhan is standing in between Atiq and Ishmam. Sabrina and Rifat are standing next to each other. Tabassum is either at the front of the line or the back of the line. Sabrina has one person behind her.

- a) Formulate the scenario as a Constraint Satisfaction Problem (CSP) by identifying the variables, values, and constraints. Also, draw the corresponding constraint graph.

9 + 5
(CO2)
(PO2)

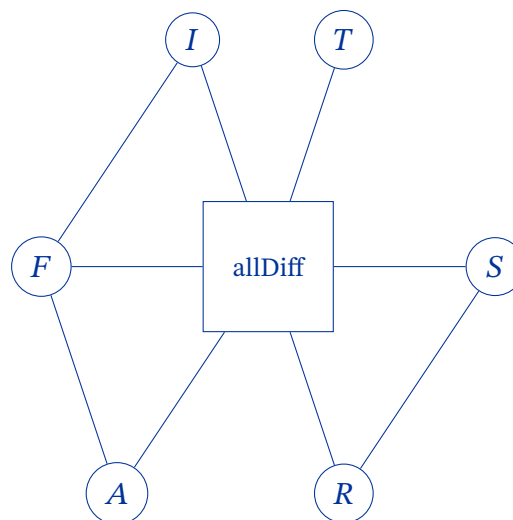
Solution:

Variables: $\{R, A, F, I, T, S\}$

Values: $\{1, 2, 3, 4, 5, 6\}$ for all variables

Constraints: $\text{allDiff}(R, A, F, I, T, S)$, $|F - A| = 1$, $|F - I| = 1$, $|S - R| = 1$, $T \in \{1, 6\}$, and $S = 5$.

The constraint graph is as follows:



Rubric:

- 0.5 points for each variable/value
- 0.5 points for each constraint
- 0.25 points for each correct node/edge
- 1 point for not including any wrong edge

- b) Consider that you apply the unary constraints and then use Arc Consistency for binary constraints to remove the values from the domains of the variables.

13
(CO1)

Determine the remaining values for each variable. You need to show the intermediate steps with a brief explanation.

(PO1)

Solution:

After applying the unary constraints, the values 2, 3, 4, 5 get removed from the domain of T , and all but 5 get removed from the domain of S .

Considering the arc $R \rightarrow S$, the values 1, 2, 3, 5 get removed from the domain of R .

No further reductions can be made using the binary constraints since A , F , and I did not lose any values.

$$R = \{4, 6\}$$

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$F = \{1, 2, 3, 4, 5, 6\}$$

$$I = \{1, 2, 3, 4, 5, 6\}$$

$$T = \{1, 6\}$$

$$S = \{5\}$$

Rubric:

- 0.25 points for each remaining/missing value of the variables
- 1 point for each explanation/decision.

3. a) Mr. Takaful's father had cancer. He knows that, even though cancer itself cannot be passed down from parents to children, there is 1% chance that the genetic change that increases the risk of cancer can be passed down (inherited). Mr. Takaful has 10M USD. He also knows that if is diagnosed with cancer, he will lose all of his money. 7 + 5
(CO3)
(PO3)

An insurance company comes to Mr. Takaful with a health insurance proposal. If he gives them an insurance premium of 5M USD, they will reimburse him 9M USD only if he gets diagnosed with cancer. Mr. Takaful's utility function is $U(x) = \ln(1 + x)$.

- i. Assume that you are a rational agent. Recommend whether Mr. Takaful should buy the health insurance or not.

Solution:

The expected utility for Mr. Takaful if he does not buy the insurance is: $0.99 \times \ln(1 + 10000000) + 0.01 \times \ln(1 + 0) \approx 15.96$

When taking the insurance, the expected utility for Mr. Takaful is: $0.99 \times \ln(1 + 5000000) + 0.01 \times \ln(1 + 5000000 + 9000000 - 10000000) \approx 15.20$

Since the expected utility is lesser, Mr. Takaful should not buy the insurance.

Rubric:

- 2 points each for expected utility
- 3 points for decision

- ii. Comment on why the insurance company is offering such a deal where they have a chance to lose 4M USD.

Solution:

The expected monetary value for the insurance is: $0.99 \times 5000000 + 0.01 \times (5000000 - 9000000) = 4910000$ USD.

Since the value is positive, the insurance company can offer the same insurance to as many people similar to Mr. Takaful as possible and gain a net profit.

Rubric:

- 2 points for the expected monetary value
- 3 points for explanation

b) Formulate the following actors as agents by identifying their sensors, actuators, and environments:

3 × 3
(CO2)
(PO2)

i. Microwave Oven

Solution:

Sensors: Button state, door-open sensor, humidity sensors, temperature sensors, etc.
Actuators: Turn on, turn off, adjust microwave power, sound completion alarm, etc.
Environment: The contents of the oven and the state of the door (open/closed)

ii. Chess Program

Solution:

Sensors: A typical program accepts keyboard, mouse, or touchscreen input indicating the opponent's intended move, resignation, hint request, etc.
Actuators: Display a legal move, or the board resulting from the move.
Environment: Representation of the board state, (for a sophisticated program) human's playing style and ability in order to improve its chances of winning or make the game more entertaining

iii. Autonomous Supply Delivery Plane

Solution:

Sensors: GPS, Air Speed, Altimeter (Air pressure sensor), sensors connected to the propellers and other actuators, 3-axis accelerometer, camera, etc.
Actuators: power to propeller, ailerons, elevators, and rudder; dropping and picking up payload
Environment: geographical area of the flight, atmospheric conditions.

c) Introspection is the process of examining or observing one's own mental and emotional processes. One of the desired characteristics of rational agents is that they need to be introspective so that can improve itself. To instill this characteristic into agents, scientists often try to understand how humans report on their own inner thoughts. But the task is difficult since it can be inaccurate.

4
(CO1)
(PO1)

Discuss how someone can be wrong about what they are thinking.

Solution:

Just as we are unaware of all the steps that go into making our heart beat, we are also unaware of most of what happens in our thoughts. We do have a conscious awareness of some of our thought process, but the majority remains opaque to our consciousness. The field of psychoanalysis is based on the idea that one needs trained professional help to analyze one's own thoughts. Neuroscience has also shown that we are unaware of much of the activity in our brains.

Rubric:

- 4 points to discussing how we are not conscious about our thought process.